required sample size. Then $n-1$ such sampling intervals each consisting of k units will be formed. A number is selected at random from the first sampling interval. Let it be number i where $i \leq k$. This number is the random starting point for the whole selection of the sample. The unit corresponding to i is the first unit in the sample. The subsequent sampling units are the units in the following positions:

$$i,\ k+i,\ 2k+i,\ 3k+i,\ \ldots,\ nk$$

The layout for systematic sampling								
Sampling interval 1	1	2	3	4	...	i - Random start	...	k
Sampling interval 2	$k+1$	$k+2$	$k+3$	$k+4$	...	$k+i$	...	2k
Sampling interval 3	$2k+1$	$2k+2$	$2k+3$	$2k+4$	...	$2k+i$	...	3k
Sampling interval 4	$3k+1$	$3k+2$	$3k+3$	$3k+4$	...	$3k+i$	...	$4k$
	...	...	...	...	...	...	...	...
Sampling interval n	$(n-1)k+1$	$(n-1)k+2$	$(n-1)k+3$	$(n-1)k+4$	...	$(n-1)k+i$	...	nk

Thus, with selection of the first unit, the whole sample is selected automatically. As the first unit could have been any of the k units, the technique will generate k systematic samples with equal probability. If N is not an integral multiple of n, then sizes of a few possible systematic samples may vary by one unit.

NOTE

This sample is also called ***quasi random sample*** since the first unit only is selected at random and all the subsequent units are not selected randomly.

Merits:

- This method is simple and convenient.
- Less time consuming.
- It can be used in infinite population.

Limitation:

- Since it is a quasi random sampling, the sample may not be a representative sample.

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